A rapid overview of systematic reviews on the effects of palm oil intake compared with intake of other vegetable oils on mortality and cardiovascular health in children and adults



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1. Introduction

Cardiovascular disease (CVD) is the leading cause of death worldwide. CVD continues its decades-long rise (in number of disability-adjusted life years [DALYs] and deaths) in most low- and middle-income countries (LMIC). The CVD burden attributable to modifiable factors also continues to increase globally (Roth 2020).

Dyslipidemia – the imbalance of lipids such as cholesterol, low-density lipoprotein (LDL), triglycerides and high-density lipoprotein (HDL) – is one of the most important modifiable risk factors for the development of CVD. In line with best available evidence and current CVD guidelines, blood concentrations of non-HDL-cholesterol and LDL-cholesterol (i.e. apolipoprotein B-containing lipoproteins) are accepted as causal cardiovascular risk factors and are a cornerstone of CVD risk prediction in the general population. Current recommendations focus on maintaining low non-HDL-cholesterol and LDL-cholesterol to reduce CVD risk (Brunner 2019; Grundy 2019; Mach 2020; Rabar 2014). Alarmingly, the total number of DALYs due to elevated LDL-cholesterol has risen steadily since 1990 (Roth 2020).

Diet is another key modifiable risk factor for CVD – the importance of dietary risks as a cause of global disease burden is undisputed (GBD 2015 Risk Factors Collaborators 2016). The effects of diet on CVD remain considerable when assessed as individual dietary factors or as measures of overall diet quality. Globally, the absolute burden of CVD attributable to dietary risks has risen for 30 years, regardless of how it is measured, and it is predicted to rise rapidly in LMIC over the coming years (Roth 2020). Many studies have shown that the relationships between different types of dietary oils or solid fats and CVD risk factors (e.g. blood lipids) can be predicted from the fatty acid composition of the former.

Palm oils are derived from the fruit of the oil palm tree, and over recent decades they have become one of the most widely produced and consumed oils in the world. Two types of oil are typically produced from oil palms: crude palm oil from the fleshy fruit and palm kernel oil from the kernel. Compared with other liquid oils, palm oil is relatively high in saturated fats. Myristic acid (1%), stearic acid (5%) and palmitic acid (44%) make up the saturated fatty acid component of palm oil (Sambanthamurthi 2000). A recent Cochrane review (*n*=15 randomized controlled trials [RCTs], *n* ~59 000 participants) reported that reducing saturated fat intake for at least 2 years causes important reductions in combined cardiovascular events, which did not change by study duration, sex or baseline CVD risk. Greater decreases in saturated fat intake caused greater decreases in cardiovascular events, and replacing the energy from saturated fat with polyunsaturated fat or carbohydrate appeared to be useful strategies; however, the effects of replacement with monounsaturated fat were less clear (Hooper 2020). Owing to the prevalent intake of palm oil in many countries and its potential to contribute to CVD risk, it is important to evaluate the available evidence on the effects on CVD risk of palm oil compared with other vegetable oils.

When seeking to better understand the available body of evidence on a particular topic, systematic reviews can be used to collectively examine the findings from all related or similar primary studies on the question of interest. Critical appraisal of the methodological quality of systematic reviews can help users to determine the quality of reviews based on how well they were conducted (Higgins 2021). In this overview, “palm oil” includes the various palm oils and fractions thereof available.

Who is this rapid overview for?

This rapid overview is aimed at people making decisions that require an understanding of the available research evidence on the effects of palm oil intake on cardiovascular health.

What does this rapid overview include and exclude?

The rapid overview includes findings from published systematic reviews, as well as assessments of the methodological quality of those reviews. It excludes assessments of the certainty of the evidence.

What is a systematic review?

A systematic review is a research study that addresses a clearly formulated question, and uses systematic and explicit methods to identify, select, appraise (i.e. assess the risk of bias) and synthesize all the primary studies that meet prespecified eligibility criteria.

What is a meta-analysis?

A meta-analysis is a statistical process to combine data from independent primary studies focused on the same question when appropriate. This analysis generates a pooled quantitative estimate of the studied phenomenon; for example, the effectiveness of an intervention.

2. Objective

The objective of this review was to rapidly identify and summarize existing evidence from published systematic reviews on the relationships between palm oil intake and mortality or cardiovascular health in children and adults, and to assess the methodological quality of these systematic reviews.

3. Methods

3.1 Inclusion and exclusion criteria

Published systematic reviews that met the criteria outlined in Table 3.1 were eligible for inclusion.

Table 3.1 Criteria for considering studies for inclusion

Item	Eligibility criteria
Population	• Children (aged 2 years and above) and adults from any country or setting.
Interventions or exposures of interest	• Interventions stating an intention to change palm oil intake by provision of foods or oils, or nutrition education in any form, or both. • Palm oil intake exposure (in grams, as a percentage of total intake, or as one of the defining characteristics of a diet or dietary pattern) comparatively analysed (e.g. lower versus higher intakes with or without replacement) in relation to eligible outcome measures over time (prospectively or retrospectively). • Excluded: — systematic reviews *only* examining other characteristics of fat intake (e.g. different types of dietary fat or total fat intake) without also including palm oil intake separately.
Comparator	• Systematic reviews comparing palm oil intake with intake of other types or categories of oils, or examining associations over time between different levels of palm oil intake and eligible outcomes.
Outcomes	• All-cause mortality • Cardiovascular mortality • Coronary heart disease (CHD) • Non-fatal myocardial infarction (MI) • Non-fatal stroke • Other “hard” cardiovascular outcomes • Routine clinical cardiovascular risk factors, such as circulating total cholesterol, LDL-cholesterol, HDL-cholesterol, non-HDL-cholesterol and triglyceride concentrations
Study design and language	• Systematic reviews assessing relationships between coconut oil intake and eligible outcomes, with no language or publication date limitations. • A systematic review will be defined as a review that had predetermined objectives, had predetermined criteria for eligibility, searched at least two data sources (one of which needed to be an electronic database), and performed data extraction and risk of bias assessment of included studies. Reviews that did not align with this definition were excluded. • Systematic reviews of qualitative or cross-sectional studies were excluded.

HDL: high-density lipoprotein; LDL: low-density lipoprotein.

3.2 Study search

In March 2021, two databases, MEDLINE (Ovid) and Epistemonikos, were searched to identify eligible systematic reviews. No date or language restrictions were applied. **Annex 1** contains the search strings per database. Another search was done in January 2022 of the databases MEDLINE (PubMed), Embase (Ovid), CINHAL, EBSCOhost, Epistemonikos, Trip Database and PROSPERO.

3.3 Study selection

After deduplication, full texts of all records from the search yield were obtained and screened against the eligibility criteria by two researchers. The reported funding and conflict of interest information in all eligible systematic reviews was extracted individually. Using this information, only systematic reviews deemed to be "independent" (i.e. without industry or potentially conflicted funding, and where authors stated no conflicts of interest) were included for further data extraction and reporting.

3.4 Critical appraisal of the methodological quality of the included systematic reviews

Undertaking a systematic review is a complex exercise and not all systematic reviews are of equal methodological quality (defined as the likelihood that the design and conduct of a systematic review will generate unbiased results). Systematic limitations in the study design, data collection, analysis and interpretation of a systematic review can bias results in ways that are becoming better understood. People producing systematic reviews are working towards preventing potential biases by developing and applying explicit expectations and recommendations for the conduct of systematic reviews.[1]

Bias can arise at all stages of the review process, and it is necessary to take potential biases into consideration when interpreting the results and conclusions of a systematic review. Critical appraisal of the methodological quality of systematic reviews can help users to determine which reviews are of high quality. A high-quality systematic review adheres to a strict scientific design based on explicit, prespecified and reproducible methods, and can provide reliable findings with defensible conclusions.

AMSTAR 2 is a tool that allows detailed critical appraisal of the methodological quality of systematic reviews of both randomized and non-randomized studies of health care interventions (Shea 2017); this tool can be used to carry out rapid and reproducible assessments. More information about AMSTAR 2, including the domains assessed, is provided in Section 4. To critically appraise the methodological quality of systematic reviews included in this overview, one researcher applied AMSTAR 2 to each relevant systematic review and a second person checked those assessments. Since AMSTAR 2 does not include domains specific to systematic reviews reporting network meta-analyses (NMA), the ISPOR-AMCP-NPC Good Practice Task Force checklist (Jansen 2014) was applied to these systematic reviews. All efforts were made to consistently and transparently apply the appraisal instruments to each review, recognizing that most assessments require a certain degree of judgement. All judgements were made based on the reporting in the full-text papers. Disagreements were resolved by discussion and final justifications for judgements were recorded. Summaries of the AMSTAR 2 or ISPOR-AMCP-NPC Good Practice Task Force assessments were produced for each domain in each included systematic review (Jansen 2014).

3.5 Data extraction and reporting

Relevant data were extracted from the "independent" eligible systematic reviews individually. Findings for the following comparisons and all eligible outcomes were extracted from and reported on:

- palm oil intake compared with other vegetable oils;

[1] The Cochrane Collaboration has formally adopted the Methodological Expectations of Cochrane Intervention Review guidelines, and the United States Institute of Medicine has recommended standards for conducting high-quality systematic reviews.

- palm oil intake compared with subgroups of vegetable oils rich in polyunsaturated fatty acids (PUFAs); and
- palm oil intake compared with subgroups of vegetable oils rich in monounsaturated fatty acids (MUFAs).

Extracted data were tabulated and described narratively, using figures where appropriate. Owing to the different methods used, findings from the systematic reviews using pairwise meta-analysis[1] were reported separately from those using NMA.[2]

3.5.1 Risk of bias assessments by authors of the systematic review

Rigorous risk of bias assessments of all the primary studies in a systematic review with a well-established and validated tool is a fundamental part of conducting a systematic review. A bias is a systematic error, or deviation from the truth, in findings or inferences. Assessment of risk of bias is an essential part of understanding the extent of credibility of results of included primary studies (Higgins 2021). For each included systematic review, the number and percentage of included primary studies that had methodological limitations were extracted, according to the risk of bias assessments done by the authors of the systematic reviews; this was reported narratively for each comparison. When systematic review authors used the Jadad scale (ranges from 0 to 5) to assess risk of bias, a cut-off value of <3 was used for "poor" in terms of risk of bias. Judgements made by systematic review authors using the Cochrane risk of bias tool for RCTs (high, unclear or low risk) or the risk of bias in non-randomized studies of interventions (ROBINS-I) (serious, moderate or low risk) did not always include an overall risk of bias judgement. In these cases, a study with at least one domain at high or serious risk was considered to be at high or serious overall risk, whereas a study with at least one domain at unclear or moderate risk, but none at high or serious risk, was considered to be at unclear or moderate overall risk. Studies assessed with these tools that only had domains at low risk of bias were judged to be at low overall risk of bias.

3.5.2 Systematic reviews reporting pairwise meta-analysis

For each eligible comparison in the systematic reviews, pooled effect estimates with their 95% confidence intervals (CI) were extracted from meta-analyses of eligible outcomes and presented in figures (forest plots), when appropriate, or reported narratively. A pooled effect estimate is an estimate of the effect of coconut oil compared with the comparator oils on a certain outcome, derived from a weighted average of effects across the included primary studies. The 95% CI is a range of two values around the pooled effect estimate, within which it is probable with 95% confidence that the true value lies, for the entire population of patients from which the study patients were selected. In theory, if the study were to be repeated 100 times under exactly the same conditions, the true value would lie within the 95% CI 95 times.

The number of participants per group and the number of studies and study designs that were combined to obtain the various pooled effect estimates, as well as the specific comparison per systematic review, were reported. Where the participants per trial arm were not reported, the following approaches were taken to estimate this: for cross-over trials, the total number of participants in the trial was used for both the intervention and control arm; for any other design, the total number of participants in the trial was divided by two, to estimate separate participant numbers in the intervention and control arm.

3.5.3 Systematic reviews reporting NMA

systematic reviews reporting NMA typically report rankings based on the cumulative probability of an intervention ranking as the best option compared with others, using the surface under the cumulative ranking curve (SUCRA). SUCRA values take on a single value between 0% and 100%, where a larger value represents a better outcome. However, these values alone do not take into account the

[1] Pairwise meta-analysis synthesizes evidence from trials comparing two interventions at a time.
[2] NMA is an extension to pairwise meta-analysis; under certain conditions, it allows for the comparison of multiple interventions in a systematic review combining direct evidence (from trials comparing two interventions) and indirect evidence (from a linked pathway via at least one comparator) simultaneously in a network of trials.

uncertainty in the relative effects that inform them; hence, it is better to report SUCRA values with 95% credible intervals (CrIs). For this review, SUCRA values were extracted and presented as percentage probabilities that each fat or oil is best for each relevant outcome.

Relative treatment effects per outcome for systematic reviews reporting NMA were extracted and summarized. Making inferences based on the probability of being ranked the best without reporting 95% CrIs, or accounting for variability in another way, is not recommended (Chaimani 2021). Since probabilities and SUCRA are very sensitive to uncertainty in the relative effects underpinning these measures, and given that these relative measures of effect may or may not be clinically meaningful, ranking should be interpreted with caution (Dias 2018, Rouse 2017).

3.5.4 Interpretation of the findings

All findings from systematic reviews should be interpreted in light of the certainty of evidence underpinning the finding. Assessment of certainty using a systematic, transparent approach such as Grading of Recommendations Assessment, Development and Evaluation (GRADE) is recommended. In GRADE, certainty refers to the extent of the confidence in the effect (the certainty that a true effect lies on one side of a specified threshold or within a chosen range) (Hultcrantz 2017). This rapid overview reports both the findings of the included systematic reviews and a critical appraisal of their methodological quality. Assessments of the certainty of the evidence were not conducted because of this being a rapid approach, and because guidance on how to adapt GRADE methodology for overviews is still under development. Instead, a transparent approach to consistently formulating the interpretations of the systematic review findings was followed. This approach is informed by the GRADE method for assessing imprecision and is described below.

Thresholds were specified for minimally important differences (MIDs) for the various outcomes when palm oil intake was compared with intakes of comparator oils. The following MID thresholds were used for the circulating lipids (continuous outcomes): total cholesterol >0.26 mmol/L, LDL-cholesterol >0.13 mmol/L, HDL-cholesterol >0.08 mmol/L and triglycerides >0.09 mmol/L. These thresholds, which represent a 5% change from the upper range of the “normal” value, were chosen because established and widely accepted MIDs are not available. Differences greater than these MID thresholds could reasonably be considered to have potential clinical relevance in terms of CVD risk, particularly on a population level, where small individual effect sizes can translate to large population health impacts; differences below the thresholds may not have clinical relevance and thus were regarded as trivial. MIDs for dichotomous outcomes were specified as a change of 25% in the event; that is, an odds ratio (OR) of <0.75 or >1.25. Such a change in a CVD event has potential clinical relevance and has been recommended as a cut-off for dichotomous outcomes (Guyatt 2011). Pooled effect estimates and 95% CIs were interpreted consistently in relation to these thresholds for continuous and dichotomous outcomes.

4. Results

4.1 Description of included systematic reviews

4.1.1 Results of the search

The study selection flowchart for the March 2021 search is available in **Annex 1 Fig. A1.1**. No additional eligible systematic reviews were found in the January 2022 search. The full texts of 38 records were assessed against the eligibility criteria and seven systematic reviews were included (Fattore 2014, Hisham 2020, Ismail 2018, Schwingshackl 2018, Sun 2015, Voon 2019, Wang 2019). Two of these systematic reviews reported conflicts related to funding, as follows:

- Fattore 2014 was partially supported by Soremartec Italia s.r.l. (a confectionery manufacturer); and
- Voon 2019 was supported by the Malaysian Palm Oil Board.

Data extraction was completed for the five remaining systematic reviews (Hisham 2020, Ismail 2018, Schwingshackl 2018, Sun 2015, Wang 2019), where no conflicts related to funding or authors were reported. Thirty-one records were excluded, with exclusion reasons summarized in **Annex 1 Fig. A1.1**, and reasons per record provided in **Annex 1 Table A1.1**.

4.1.2 Included systematic reviews

Four of the five systematic reviews conducted pairwise meta-analysis (Hisham 2020, Ismail 2018, Sun 2015, Wang 2019) and the other conducted an NMA (Schwingshackl 2018). The characteristics of the five included systematic reviews are summarized in **Annex 2 Table A2.1**. This table includes each review's aim or objective; eligible primary study designs, participants and comparison; duration of intervention; prespecified outcomes; number of primary studies and study designs included; countries where the primary studies were done; and reported funding information and conflict of interest data.

Eligible participants

Three systematic reviews (Hisham 2020, Sun 2015, Wang 2019) did not specify age as an inclusion criterion, but only adults were eligible in the other two systematic reviews (Ismail 2018, Schwingshackl 2018). Wang 2019 additionally specified that eligible participants should be free of any metabolic syndrome or medical history of coronary artery disease, and should not be taking any form of medicine such as oral contraceptives.

Eligible comparisons

Four of the systematic reviews sought to include studies that compared palm oil or fat intake with intake of any other type of oil or fat. Two systematic reviews included studies comparing oils only, one included studies comparing cooked foods or oils, and another included studies comparing diets rich in the oils of interest. Schwingshackl 2018 included studies comparing diets varying in composition or isocaloric exchange of at least two of the following: coconut, corn, flaxseed, hempseed, olive, palm, rapeseed, safflower, soybean and sunflower oils; and beef fat, butter and lard.

Outcomes

The included reviews only reported effects on the following outcomes eligible for this summary: circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides; and non-fatal first acute myocardial infarction (MI).

Duration

Included trials in four systematic reviews ranged from 2 to 27 weeks; included case-controls in the remaining systematic review ranged from 4 to 18 years.

4.1.3 Critical appraisal of the included systematic reviews (methodological quality)

Using AMSTAR 2, Hisham 2020 was judged as being a "low quality" systematic review; Ismail 2018, Sun 2015 and Wang 2019 were judged to be "critically low quality" systematic reviews. A key limitation of Sun 2015 and Wang 2019 was the pooling of data from RCTs and non-randomized trials. Ismail 2018 did not include any data from intervention studies and did not synthesize evidence from included studies. **Annex 2 Table A2.2** contains guidance for interpreting the AMSTAR 2 ratings of overall confidence in the results of a systematic review. For these ratings, AMSTAR 2 identifies critical weaknesses that should reduce confidence in the findings of a systematic review; these are described in **Annex 2 Table A2.3**. See **Annex 2 Table A2.4** for AMSTAR 2 judgements per included systematic review.

Using the ISPOR-AMCP-NPC Good Practice Task Force checklist, the credibility of the NMA by Schwingshackl 2018 was judged as being "sufficient". Key non-critical limitations identified in the appraisal of Schwingshackl 2018 were the presence of effect modifiers that were not accounted for in the analysis, and heterogeneity that was not explored with subgroups or meta-regression. See **Annex 2 Table A2.5** for the ISPOR-AMCP-NPC Good Practice Task Force checklist judgements for the NMA by Schwingshackl 2018.

4.2 Effects of palm oil intake – risk of bias assessments

4.2.1 Palm oil intake compared directly with other vegetable oils in pairwise meta-analyses: circulating blood lipids

For this comparison, 21 primary studies (84%) included in Sun 2020 were judged as being "poor" in terms of risk of bias (scoring <3 on the Jadad scale); 11 primary studies (100%) included in Wang 2019 were at high overall risk of bias using the Cochrane risk of bias tool.

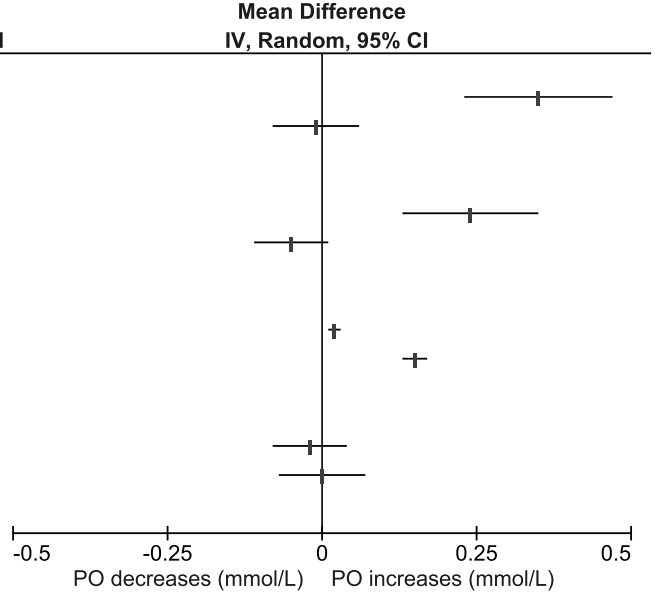
Fig. 4.1 shows the pooled effect sizes (mean difference [MD]) and 95% CIs for circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides when palm oil intake was compared with other vegetable oils by Sun 2015 and Wang 2019. Importantly, the comparisons in these reviews were distinctly different. Sun 2015 compared palm oil intake with the intake of vegetable oils low in saturated fat (*n*=27 studies) while Wang 2019 compared a diet rich in palm oil with a diet rich in unsaturated fatty acids, and included fewer studies (*n*=12 studies, 11 of which were also included in Sun 2015).

The MD and 95% CI for Sun 2015 indicated that, on average, palm oil intake compared with the intake of other vegetable oils may be associated with a potential clinically relevant effect (0.35 mmol/L) on total cholesterol (*n*=1407), whereas Wang 2019 indicated that a diet rich in palm oil compared with a diet rich in unsaturated fatty acids is probably associated with a trivial effect on total cholesterol (*n*=961). Sun 2015 indicated that the intake of palm oil compared with vegetable oils low in saturated fat may be associated with a potential clinically relevant effect (0.24 mmol/L) on LDL-cholesterol (*n*=1379), on average. However, Wang 2019 reported that a diet rich in palm oil compared with a diet rich in unsaturated fatty acids is probably associated with a trivial effect on LDL-cholesterol (*n*=961), on average.

Sun 2015 indicated that the intake of palm oil compared with vegetable oils low in saturated fat is probably associated with a trivial effect on HDL-cholesterol (*n*=1369), on average. For HDL-cholesterol, Wang 2019 reported that a diet rich in palm oil compared with a diet rich in unsaturated fatty acids is probably associated with a potential clinically relevant effect (0.15 mmol/L), on average (*n*=767). Both systematic reviews indicated that palm oil intake is probably associated with a trivial effect on triglycerides, on average (Sun 2015, *n*=1341; Wang 2019, *n*=741).

Fig. 4.1 Pooled effect sizes (95% CI) of palm oil intake compared with other vegetable oils on circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides, reported per included systematic review

Study or Subgroup	Mean Difference	SE	Experimental Total	Control Total	Mean Difference IV, Random, 95% CI
1.1.1 Total cholesterol					
Sun 2015 (1)	0.35	0.0612	704	703	0.35 [0.23, 0.47]
Wang 2019 (2)	-0.01	0.0357	481	480	-0.01 [-0.08, 0.06]
1.1.2 LDL-cholesterol					
Sun 2015 (3)	0.24	0.0561	690	689	0.24 [0.13, 0.35]
Wang 2019	-0.05	0.0306	481	480	-0.05 [-0.11, 0.01]
1.1.3 HDL-cholesterol					
Sun 2015 (4)	0.02	0.0051	685	684	0.02 [0.01, 0.03]
Wang 2019	0.15	0.0102	370	397	0.15 [0.13, 0.17]
1.1.4 Triglycerides					
Sun 2015 (5)	-0.02	0.0306	671	670	-0.02 [-0.08, 0.04]
Wang 2019	0	0.0357	370	371	0.00 [-0.07, 0.07]



When not reported, totals per group were estimated from study totals

Footnotes
(1) 27 studies (crossover, randomized crossover, parallel and randomized parallel trials)
(2) 12 studies for all outcomes (crossover, randomized crossover and randomized parallel trials; before-after study)
(3) 26 studies (crossover, randomized crossover, parallel and randomized parallel trials)
(4) 26 studies (crossover, randomized crossover, parallel and randomized parallel trials)
(5) 25 studies (crossover, randomized crossover, parallel and randomized parallel trials)

Specific comparisons per review
Sun 2015: Palm oil vs vegetable oil low in saturated fat
Wang 2019: Diet rich in palm oil vs diet rich in unsaturated fatty acids

CI: confidence interval; HDL: high-density lipoprotein; IV: inverse variance; LDL: low-density lipoprotein; PO: palm oil; SE: standard error; TC: total cholesterol.

4.2.2 Palm oil intake compared directly with other vegetable oils in pairwise meta-analyses: non-fatal first acute MI

For this comparison, a single primary study included in Ismail 2018 that reported on this outcome was judged to have a low overall risk of bias using the Critical Appraisal Skills Programme (CASP) tool.

Fig. 4.2 shows the pooled effect size (OR) and 95% CI for non-fatal first acute MI, diagnosed using WHO criteria, reported by one systematic review (Ismail 2018). This systematic review indicated that palm oil intake compared with the intake of vegetable oils may be associated with an increase in non-fatal first acute MI, on average (*n*=2091).

4.2.3 Palm oil intake compared directly with PUFA-rich vegetable oils in pairwise meta-analyses: circulating blood lipids

For this comparison, eight primary studies (100%) included in Hisham 2020 were at unclear risk of overall bias using the Cochrane risk of bias tool; six studies (100%) included in Wang 2019 were at high risk of overall bias using the same tool.

Fig. 4.3 shows the MD and 95% CI for circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides when palm oil intake was compared with PUFA-rich vegetable oils by Hisham 2020 and Wang 2019 (for HDL-cholesterol and triglycerides only). Importantly, Hisham 2020 reported results from cross-over trials (*n*=228) separately from results of parallel trials (*n*=113 to 152 across outcomes).

Hisham 2020 reported that palm oil intake compared with PUFA-rich vegetable oils may be associated with a potential clinically relevant effect on total cholesterol, on average; cross-over trials (*n*=228) and parallel trials (*n*=152) both indicated effects of potential clinical relevance (0.37 mmol/L and 0.32 mmol/L, respectively). On average, evidence from cross-over trials (*n*=228; Hisham 2020) showed that the intake of palm oil compared with PUFA-rich vegetable oils may be associated with a trivial effect on LDL-cholesterol. Data from parallel trials (*n*=152; Hisham 2020) indicated that the intake of palm oil may be associated with increased LDL-cholesterol, on average; however, the 95% CI for this MD was very wide.

MDs and 95% CIs reported by Hisham 2020 (cross-over, *n*=228; parallel trials, *n*=152) and Wang 2019 (*n*=311) indicated that palm oil intake may be associated with a trivial effect on HDL-cholesterol. Hisham 2020 (cross-over, *n*=228; parallel trials, *n*=113) indicated that palm oil intake may be associated with a trivial effect on triglycerides, on average. The effect reported by Wang 2019 was slightly more precise, but also suggested that a diet rich in palm oil compared with a diet rich in PUFA may be associated with a trivial effect on triglycerides.

4.2.4 Palm oil intake compared directly with MUFA-rich vegetable oils in pairwise meta-analyses: circulating blood lipids

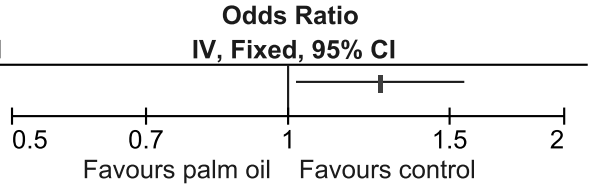
For this comparison, 15 primary studies (93.75%) included in Hisham 2020 were at unclear risk of overall bias using the Cochrane risk of bias tool; the remaining included study (6%) was at high risk of overall bias using the same tool. Five studies (100%) included in Wang 2019 were at high risk of overall bias, also using the Cochrane risk of bias tool.

Fig. 4.4 shows the MDs and 95% CI for circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides when palm oil intake was compared with MUFA-rich vegetable oils by Hisham 2020 and Wang 2019 (HDL-cholesterol and triglycerides only). Hisham 2020 also reported results from cross-over trials (*n*=730) and a single parallel trial (*n*=60) separately for the comparison of palm oil intake with MUFA-rich vegetable oils. Evidence from the single small parallel trial is very uncertain for all the circulating blood lipids, and was probably affected by small study bias.

Results from cross-over trials included in Hisham 2020 (*n*=730) indicated that palm oil intake compared with MUFA-rich vegetable oils may be associated with a potential clinically relevant effect (0.31 mmol/L) on total cholesterol, on average. For LDL-cholesterol, evidence from cross-over trials (*n*=730, Hisham 2020) showed that the intake of palm oil compared with MUFA-rich vegetable oils may be associated with a potential clinically relevant effect (0.24 mmol/L), on average.

Fig. 4.2 Pooled effect sizes (95% CI) of palm oil intake compared with other vegetable oils on non-fatal MI, reported per included systematic review

Study or Subgroup	log[Odds Ratio]	SE	Cases Total	Controls Total	Odds Ratio IV, Fixed, 95% CI
Ismail 2018 (1)	0.2311	0.1078	1077	1014	1.26 [1.02, 1.56]



Footnotes

(1) 1 study (case-control)

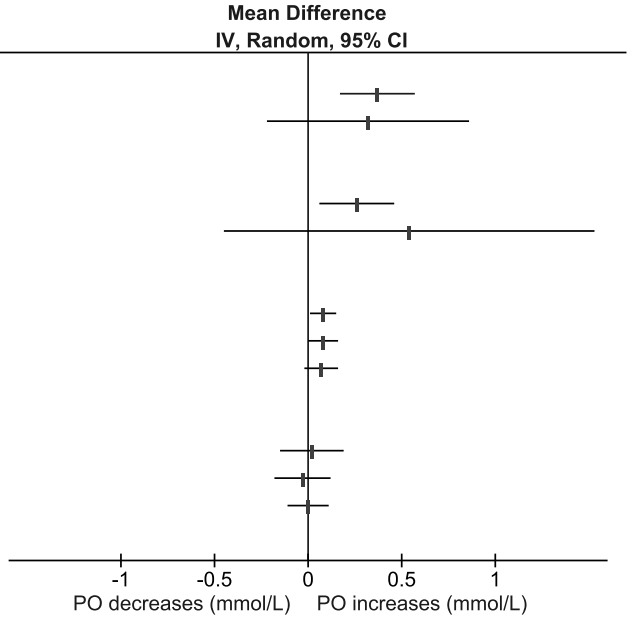
Specific comparison in the review

Ismail 2018: Palm oil vs vegetable oils

CI: confidence interval; IV: inverse variance; MI: myocardial infarction; SE: standard error.

Fig. 4.3 Pooled effect sizes (95% CI) of palm oil intake compared with PUFA-rich vegetable oils on circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides, reported per included systematic review

Study or Subgroup	Mean Difference	SE	Experimental Total	Control Total	Mean Difference IV, Random, 95% CI
3.1.1 Total cholesterol					
Hisham 2020 (crossover) (1)	0.37	0.102	114	114	0.37 [0.17, 0.57]
Hisham 2020 (parallel) (2)	0.32	0.2755	76	76	0.32 [-0.22, 0.86]
3.1.2 LDL-cholesterol					
Hisham 2020 (crossover)	0.26	0.102	114	114	0.26 [0.06, 0.46]
Hisham 2020 (parallel) (3)	0.54	0.5051	76	76	0.54 [-0.45, 1.53]
3.1.3 HDL-cholesterol					
Hisham 2020 (crossover)	0.08	0.0357	114	114	0.08 [0.01, 0.15]
Hisham 2020 (parallel) (4)	0.08	0.0408	76	76	0.08 [0.00, 0.16]
Wang 2019 (5)	0.07	0.0459	142	169	0.07 [-0.02, 0.16]
3.1.4 Triglycerides					
Hisham 2020 (crossover)	0.02	0.0867	114	114	0.02 [-0.15, 0.19]
Hisham 2020 (parallel) (6)	-0.03	0.0765	56	57	-0.03 [-0.18, 0.12]
Wang 2019	0	0.0561	142	143	0.00 [-0.11, 0.11]



When not reported, totals per group were estimated from study totals

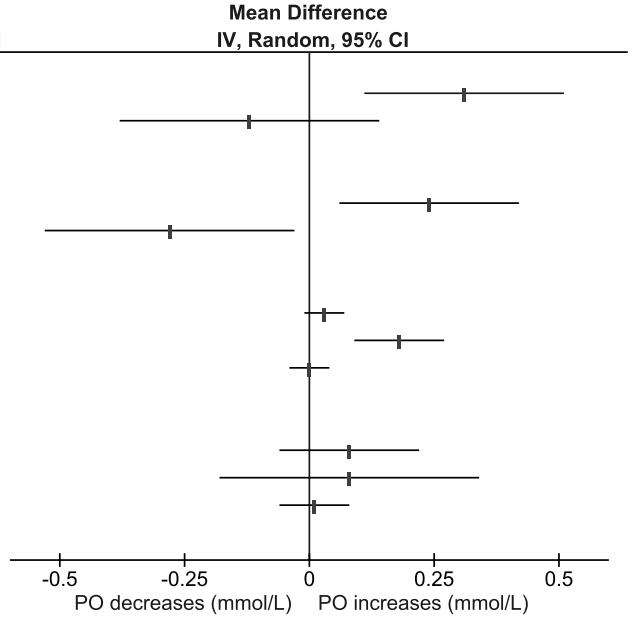
Footnotes
(1) 5 studies for all outcomes (crossover trials)
(2) 3 studies (parallel trials)
(3) 3 studies (parallel trials)
(4) 3 studies (parallel trials)
(5) 7 studies for both outcomes (crossover, randomized crossover and randomized parallel trials; before-after study)
(6) 2 studies (parallel trials)

Specific comparisons per review
Hisham 2020: Palm oil/olein vs PUFA-rich vegetable oil
Wang 2019: Diet rich in palm oil vs diet rich in PUFA

CI: confidence interval; HDL: high-density lipoprotein; IV: inverse variance; LDL: low-density lipoprotein; PO: palm oil; PUFA: polyunsaturated fatty acids; SE: standard error.

Fig. 4.4 Pooled effect sizes (95% CI) of palm oil intake compared with MUFA-rich vegetable oils on circulating total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides, reported per included systematic review

Study or Subgroup	Mean Difference	SE	Experimental Total	Control Total	Mean Difference IV, Random, 95% CI
2.1.1 Total cholesterol					
Hisham 2020 (crossover) (1)	0.31	0.102	365	365	0.31 [0.11, 0.51]
Hisham 2020 (parallel) (2)	-0.12	0.1327	30	30	-0.12 [-0.38, 0.14]
2.1.2 LDL-cholesterol					
Hisham 2020 (crossover)	0.24	0.0918	365	365	0.24 [0.06, 0.42]
Hisham 2020 (parallel)	-0.28	0.1276	30	30	-0.28 [-0.53, -0.03]
2.1.3 HDL-cholesterol					
Hisham 2020 (crossover)	0.03	0.0204	365	365	0.03 [-0.01, 0.07]
Hisham 2020 (parallel)	0.18	0.0459	30	30	0.18 [0.09, 0.27]
Wang 2019 (3)	0	0.0204	228	228	0.00 [-0.04, 0.04]
2.1.4 Triglycerides					
Hisham 2020 (crossover)	0.08	0.0714	365	365	0.08 [-0.06, 0.22]
Hisham 2020 (parallel)	0.08	0.1327	30	30	0.08 [-0.18, 0.34]
Wang 2019	0.01	0.0357	228	228	0.01 [-0.06, 0.08]



When not reported, totals per group were estimated from study totals

Footnotes
(1) 16 studies for all outcomes (crossover trials)
(2) 1 study for all outcomes (parallel trial)
(3) 5 studies for both outcomes (randomized crossover and randomized parallel trials)

Specific comparisons per review
Hisham 2020: Palm oil/olein vs MUFA-rich vegetable oil
Wang 2019: Diet rich in palm oil vs diet rich in MUFA

CI: confidence interval; HDL: high-density lipoprotein; IV: inverse variance; LDL: low-density lipoprotein; MUFA: monounsaturated fatty acids; PO: palm oil; SE: standard error.

MDs and 95% CIs reported by cross-over trials included in Hisham 2020 (*n*=730) and studies included by Wang 2019 (*n*=456) indicated that palm oil intake is probably associated with a trivial effect on HDL-cholesterol. Hisham 2020 (cross-over trials, *n*=730) indicated that palm oil intake may be associated, on average, with a trivial effect on triglycerides when compared with the intake of MUFA-rich vegetable oil. The pooled effect size reported by Wang 2019 (*n*=456) was more precise, but also suggested that a diet rich in palm oil compared with a diet rich in MUFA may be associated with a trivial effect on triglycerides.

4.2.5 Palm oil intake compared directly and indirectly with individual fats and oils in an NMA

Three primary studies (6%) included in the NMA by Schwingshackl 2018 were judged to be at high overall risk of bias and 18 (33%) were at unclear overall risk of bias using the Cochrane risk of bias tool; the remaining studies (61%) were at low overall risk of bias.

The NMA included 54 RCTs with 2065 participants, and conducted a simultaneous comparison of 13 individual fats and oils (three types of animal fat and 10 different plant oils, including palm oil) through an extension of pairwise meta-analysis. This allowed comparisons of more than two interventions in a single analysis, and also allowed the ranking of all 13 fats and oils in relation to each other.

Table 4.1 shows the cumulative probabilities – using SUCRA, – of each fat or oil included in the NMA being the best for reducing total cholesterol, LDL-cholesterol and triglycerides, and increasing HDL-cholesterol, when compared with all the other fats and oils (Schwingshackl 2018). An important limitation in the NMA was that reported SUCRA values were not accompanied by 95% CrIs; thus, these probabilities should be interpreted with caution. Compared with all other vegetable oils, palm oil had the second-lowest cumulative probability for reducing total cholesterol, and it was tied with coconut oil for the lowest cumulative probability for LDL-cholesterol. Palm oil had the second-highest cumulative probability, after coconut oil, for increasing HDL-cholesterol and the highest cumulative probability of reducing triglycerides.

The rankings by SUCRA reported by Schwingshackl should be interpreted in light of the relative effects per outcome when palm oil was compared with the other oils or fats. These relative effects (MDs and 95% CIs) are summarized in Table 4.2 below. The ranges of values for the 95% CIs of the reported MDs varied for all outcomes across the different comparisons; however, general trends of the MDs were mostly indicative of trivial increases in total cholesterol, LDL-cholesterol, HDL-cholesterol and triglycerides, when palm oil was compared with each of the other nine vegetable oils (Table 4.2).

Sensitivity analyses by Schwingshackl 2018 restricting studies to those at low risk of bias, those where investigators provided oils and fats, and those in healthy participants only did not meaningfully change the results of the primary analysis. Restriction of studies to only those at low risk of bias also did not change the results of the primary analysis significantly; however, effects on total cholesterol differed slightly in the sensitivity analysis, with larger increases on average observed for palm oil compared with other oils or fats (Schwingshackl 2018).

Several other methodological factors need to be considered when looking at the data from the NMA (Schwingshackl 2018). Significant incoherence (analogous to inconsistency in pairwise meta-analysis) was suggested for LDL-cholesterol (p=0.010) and HDL-cholesterol (p=0.009) in comparisons of palm oil with soybean oil when using the side-splitting approach. The loop-specific approach to assessing coherence identified no important incoherence between direct and indirect evidence in the loops containing palm oil (Schwingshackl 2018).

Table 4.1 NMA cumulative probabilities (%) using SUCRA of each fat or oil being best for a specific outcome

NMA probabilities of each fat or oil being best for reducing total cholesterol	
Fat or oil	**SUCRA**
Safflower oil	90
Rapeseed oil	85
Corn oil	72
Sunflower oil	72
Hempseed oil	61
Flaxseed oil	59
Soybean oil	59
Olive oil	43
Beef fat	41
Palm oil	34
Coconut oil	22
Lard	11
Butter	3

NMA probabilities of each fat or oil being best for reducing LDL-cholesterol	
Fat or oil	**SUCRA**
Safflower oil	82
Rapeseed oil	76
Flaxseed oil	71
Sunflower oil	71
Hempseed oil	69
Corn oil	66
Beef fat	50
Soybean oil	50
Olive oil	37
Coconut oil	33
Palm oil	33
Lard	10
Butter	2

NMA probabilities of each fat or oil being best for increasing HDL-cholesterol	
Fat or oil	**SUCRA**
Coconut oil	88
Palm oil	80
Beef fat	74
Hempseed oil	59
Sunflower oil	57
Lard	55
Rapeseed oil	53
Olive oil	52
Flaxseed oil	47
Butter	37
Corn oil	29
Soybean oil	13
Safflower oil	6

NMA probabilities of each fat or oil being best for reducing triglycerides	
Fat or oil	**SUCRA**
Palm oil	74
Soybean oil	72
Safflower oil	68
Corn oil	66
Hempseed oil	63
Sunflower oil	61
Rapeseed oil	58
Flaxseed oil	56
Lard	50
Olive oil	32
Coconut oil	29
Butter	17
Beef fat	6

CrI: credible interval; HDL: high-density lipoprotein; LDL: low-density lipoprotein; NMA: network meta-analysis; SUCRA: surface under the cumulative ranking curve.

Notes: A greater SUCRA is associated with a greater probability of the fat or oil being best for the outcome. Reported SUCRA values were not accompanied by 95% CrIs and reported probabilities should be interpreted with caution.

Source: Schwingshackl 2018.

Table 4.2 Relative treatment effect of oils or fats and palm oil per outcome; mean difference (95% CI) mmol/L

Palm oil versus	Total cholesterol	LDL-cholesterol	HDL-cholesterol	Triglycerides
Butter	−0.25 (−0.36 to −0.13)	−0.24 (−0.36 to −0.12)	0.03 (0.00 to 0.06)	−0.06 (−0.10 to −0.01)
Lard	−0.17 (−0.38 to 0.04)	−0.16 (−0.34 to 0.03)	0.02 (−0.02 to 0.05)	−0.02 (−0.12 to 0.07)
Beef fat	0.01 (−0.18 to 0.20)	0.04 (−0.17 to 0.25)	−0.00 (−0.05 to 0.05)	−0.09 (−0.15 to −0.03)
Coconut oil	−0.07 (−0.22 to 0.09)	−0.01 (−0.16 to 0.15)	−0.01 (−0.04 to 0.02)	−0.04 (−0.08 to −0.01)
Soybean oil	0.08 (−0.01 to 0.17)	0.04 (−0.05 to 0.13)	0.05 (0.03 to 0.07)	−0.00 (−0.05 to 0.04)
Olive oil	0.03 (−0.05 to 0.12)	0.01 (−0.07 to 0.10)	0.02 (0.00 to 0.04)	−0.04 (−0.08 to 0.00)
Corn oil	0.13 (0.02 to 0.24)	0.09 (−0.02 to 0.20)	0.04 (0.01 to 0.06)	−0.01 (−0.07 to 0.05)
Flaxseed oil	0.08 (−0.15 to 0.32)	0.12 (−0.10 to 0.35)	0.03 (−0.03 to 0.08)	−0.02 (−0.11 to 0.07)
Hempseed oil	0.11 (−0.33 to 0.54)	0.16 (−0.31 to 0.63)	0.00 (−0.20 to 0.21)	0.00 (−0.17 to 0.18)
Rapeseed oil	0.18 (0.05 to 0.32)	0.12 (−0.01 to 0.26)	0.02 (−0.02 to 0.06)	−0.02 (−0.07 to 0.04)
Sunflower oil	0.13 (0.04 to 0.21)	0.10 (0.02 to 0.19)	0.02 (−0.00 to 0.04)	−0.01 (−0.05 to 0.03)
Safflower oil	0.24 (0.03 to 0.46)	0.18 (−0.05 to 0.41)	0.08 (0.03 to 0.12)	−0.00 (−0.07 to 0.06)

CI: confidence interval; HDL: high-density lipoprotein; LDL: low-density lipoprotein.

Note: Relative effects were calculated as mean (palm oil) minus mean (comparator).

Source: Schwingshackl 2018.

5. Discussion

Five systematic reviews (one NMA) were included in this rapid overview and findings from 11–54 interventional (randomized, non-randomized, controlled trials, parallel and cross-over designs) and four observational studies (case–control and retrospective ecological) in adults, with durations ranging from 2 to 27 weeks for systematic reviews including interventional designs (Hisham 2020, Schwingshackl 2018, Sun 2015, Wang 2019) and ranging from 4 to 18 years for a systematic review including observational designs (Ismail 2018) were synthesized. Using AMSTAR 2, the methodological quality of Hisham 2020 was judged as being "low", and Ismail 2018, Sun 2015 and Wang 2019 were judged as being "critically low quality" systematic reviews. Using the ISPOR-AMCP-NPC Good Practice Task Force checklist, the credibility of the NMA by Schwingshackl 2018 was judged as being "sufficient". Using different risk of bias tools, most of the systematic review authors judged that 84–100% of primary studies reporting on circulating lipid levels had some risk of bias. The authors of the systematic review including a primary study reporting on non-fatal first acute MI judged that this study did not have concerns related to bias.

5.1 LDL-cholesterol, total cholesterol and triglycerides

The two systematic reviews reported variable effects on total cholesterol (MD range −0.01 to 0.35 mmol/L), LDL-cholesterol (MD range −0.05 to 0.24 mmol/L) and HDL-cholesterol (MD range 0.02 to 0.15) for the direct comparison of palm oil intake with the intake of other vegetable oils. This may be because of the distinctly different specific comparisons in each review that resulted in the inclusion of much fewer eligible studies by Wang 2019 (*n*=12) compared with Sun 2015 (*n*=27). Pooled effects in these systematic reviews were variable, showing that palm oil intake is probably associated with trivial effects on total cholesterol, on average, or may be associated with a potential clinically relevant effect. This was also the case for pooled effects on LDL-cholesterol reported by the systematic reviews. Pooled direct evidence for effects on triglycerides when palm oil intake was compared with the intake of other vegetable oils also indicated probable trivial effects, on average. Mean effects reported by the systematic review using NMA largely mirrored the directions of these findings for total cholesterol, LDL-cholesterol and triglycerides, when palm oil was compared with other vegetable oils, although MDs were mostly smaller (Schwingshackl 2018). Findings from the systematic reviews comparing palm oil intake with the intake of subgroups of PUFA-rich vegetable oils reported that, on average, the intake of palm oil may be associated with effects on LDL-cholesterol and triglycerides that are trivial, but it may be associated with potential clinically relevant effects on total cholesterol. Findings from the systematic reviews comparing palm oil intake with the intake of subgroups of MUFA-rich vegetable oils indicated that, on average, the intake of palm oil may be associated with effects on total cholesterol and LDL-cholesterol that are potentially clinically relevant, but it may be associated with effects on triglycerides that are trivial.

Overall, these systematic reviews report that effects of palm oil intake compared with other vegetable oils on total cholesterol may be clinically meaningful but effects on triglycerides are probably not clinically meaningful. Notably, pooled effect sizes for LDL-cholesterol reported by the largest systematic review in each of the three comparisons were consistently above the threshold for a potential clinically relevant effect in terms of CVD risk (>0.13 mmol/L). Each 0.26 mmol/L increase in LDL-cholesterol has been associated with a 14% increased risk of atherosclerosis in adults without conventional cardiovascular risk factors (Fernandez-Friera 2017). Current recommendations in Europe

and the United States of America focus on maintaining low non-HDL-cholesterol and LDL-cholesterol to reduce CVD risk (Brunner 2019; Grundy 2019; Mach 2020; Rabar 2014).

The role of LDL-cholesterol as a cause of atherosclerotic disease is well established. This has been repeatedly demonstrated in experimental studies, epidemiological cohorts, randomized clinical trials of LDL-cholesterol-lowering drugs, and studies using Mendelian randomization, where effects of low LDL-cholesterol on CVD risk can be determined, independent of confounding variables such as diet, medical therapy and weight (Ference 2017). Importantly, the gradient of the relationship between LDL-cholesterol and atherosclerotic CVD outcomes becomes steeper with increasing duration of follow-up in epidemiological studies and in intervention trials. From this, it can be concluded that an individual's risk of atherosclerotic disease is strongly determined by their cumulative lifelong LDL-cholesterol exposure (Ference 2017), and data show that lower LDL-cholesterol across the lifespan is most protective against CVD. Accordingly, a significant long-term increased risk for coronary heart disease (CHD) and cardiovascular mortality has been reported in young adults with LDL-cholesterol of 2.5 mmol/L or more (Abdullah 2018 [*n*=36 375, median age 42, median follow-up 26.8 years]; Zhang 2019 [*n*=36 030, young=18–39 years, median follow-up 17 years]). In terms of baseline risk, a recent meta-analysis evaluating the safety and effectiveness of lowering LDL-cholesterol revealed a consistent relative risk reduction in major cardiovascular events per change in LDL-cholesterol in patient populations starting at levels as low as a median 1.6 mmol/L (Sabatine 2018).

5.2 HDL-cholesterol

Pooled direct evidence from the included systematic reviews suggested variable effects of palm oil intake, with one systematic review showing that, on average, it is probably associated with a potential clinically relevant effect and another showing that, on average, it is probably associated with a trivial effect. The same direction of effects on HDL-cholesterol were seen across systematic reviews when palm oil intake was compared directly with subgroups of PUFA-rich vegetable oils and MUFA-rich vegetable oils, but comparisons of palm oil intake with subgroups of MUFA-rich vegetable oils indicated probable trivial effects whereas comparisons with PUFA-rich vegetable oils resulted in less certainty around whether the effect was trivial. Results reported by the systematic review using NMA largely mirrored the directions of these findings when palm oil was compared with other vegetable oils, but MDs were even smaller with slightly more variation around these pooled effect estimates (Schwingshackl 2018). The trivial increases in circulating HDL-cholesterol reported by systematic reviews comparing palm oil intake with various other vegetables oils are probably not clinically important in terms of CVD risk.

Although HDL-cholesterol is often referred to as the "good cholesterol", the earlier understanding that higher levels of HDL-cholesterol are protective against CVD has been challenged by recent evidence from genetic, epidemiological and intervention studies. Results from genetic studies and meta-analysis of pharmaceutical intervention trials have failed to show that modifying HDL-cholesterol changes CVD event risk (Vitali 2017, Keene 2014), despite the inverse association between HDL-cholesterol and CVD event risk seen in some observational studies. Large-scale prospective cohort studies contradict the previous finding of a linear inverse relationship between HDL-cholesterol and CVD, revealing a plateau in the inverse association above certain HDL-cholesterol, and even possible harm at extremely high levels (Emerging Risk Factors Collaboration 2009, Wilkins 2014, Madsen 2017, Bowe 2016). Mendelian randomization studies have consistently shown that raised HDL-cholesterol levels resulting from common variants in HDL-related genes are not necessarily associated with a lower incidence of cardiovascular events (Voight 2012, Haase 2012). Concentration of HDL-cholesterol represents a static picture of a dynamic and heterogeneous metabolic milieu, and current genetic data indicate it is unlikely to be itself causally protective against CHD (Vitali 2017). Considered together, current data indicate that higher HDL-cholesterol is not necessarily protective against CVD, and may even be harmful when extremely high, although this finding has yet to be confirmed. Current European dyslipidaemia guidelines emphasize that the risk of atherosclerotic CVD appears to increase when HDL-cholesterol is above 2.3 mmol/L (Mach 2020). Non-HDL-cholesterol concentrations are currently recommended in Europe and the United States for CVD risk prediction in the general population

(Grundy 2019; Mach 2020; Rabar 2014), with evidence showing strong associations with long-term risk of atherosclerotic CVD (Brunner 2019 [*n*=398 846, median age 51 years, median follow-up 13.5 years]).

5.3 Non-fatal MI

Pooled direct evidence from one study (follow-up 9 years; *n* ~2000) included in a systematic review and judged as being at low overall risk of bias indicated that palm oil intake may be associated with an increase in the odds of having a non-fatal first acute MI, on average, relative to intake of other vegetable oils. The reported odds were 1.26, with the 95% CI for this effect not crossing the line of no difference.

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Annex 1

Search strings and study selection flowchart

Searches were run for coconut oil and palm oil at the same time, and relevant records were split per topic during the full-text screening for the preparation of separate reports per topic.

Database: **Ovid MEDLINE** and Epub Ahead of Print, In-Process, In-Data-Review & Other Non-Indexed Citations, Daily and Versions <1946 to March 09, 2021>

Search strategy:

1 (coconut adj3 oil*).ti,ab. (2243)

2 (palm adj oil*).ti,ab. (2536)

3 exp Coconut Oil/ (806)

4 exp Palm Oil/ (1732)

5 1 or 2 or 3 or 4 (5348)

6 exp meta-analysis/ (127 818)

7 (systematic adj review*).ti,ab. (196 419)

8 ((meta adj anal*) or meta-anal*).ti,ab. (194 439)

9 6 or 7 or 8 (320 361)

10 5 and 9 (27)

Yield: 20

Database: Epistemonikos

n=37

Search strategy:

(title:((title:(palm oil*) OR abstract:(palm oil*)) OR (title:(coconut oil*) OR abstract:(coconut oil*))) OR abstract:((title:(palm oil*) OR abstract:(palm oil*)) OR (title:(coconut oil*) OR abstract:(coconut oil*))))

Filter: Systematic reviews

Yield: 37

Fig. A1.1 Study selection flowchart with the number of records identified, included and excluded

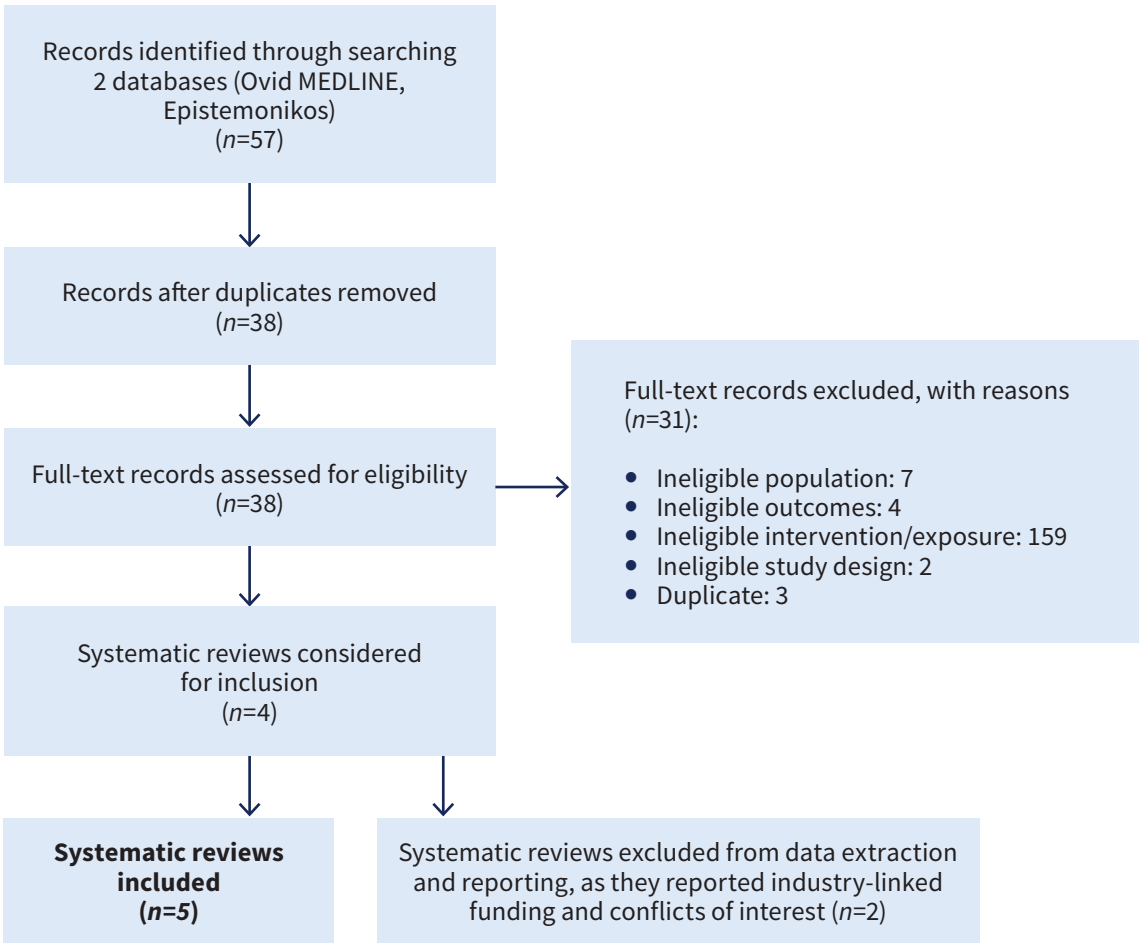


Table A1.1 Excluded records with reasons (*n*=31)

Reference	Oil type	Reason for exclusion
Berraaouan 2013	Soybean, argan, olive, palm, walnut, black cumin, safflower, colocynth, black seed, rice bran, cinnamon and rocket oil	Ineligible outcomes: diabetes
Bittar 2017	Coconut oil	Ineligible intervention/exposure: coconut oil
Boyd 2019	Herbal medicinal products (e.g. nutmeg preparation, with mace, nutmeg and coconut oil)	Ineligible intervention/exposure: coconut oil included in nutmeg analgesic preparation
Bronsky 2013	Palm oil	Ineligible study design: position paper
Daley 2020	N/A	Ineligible population: animal study
Dong 2017	Red palm oil	Ineligible outcomes: vitamin A deficiency
Eyres 2016	Coconut oil	Ineligible intervention/exposure: coconut oil
Ezekwe 2020	Argan oil, black castor oil, chamomile oil, coconut oil, horsetail plant oil, jojoba oil, lavender oil, menthol olive oil, peppermint oil, pumpkin seed oil, rosemary oil, sulfur oil, tea tree oil and thyme oil	Ineligible intervention/exposure: topical applications of oils
Fattore 2015	N/A	Original language duplicate of Fattore 2014
Fattore 2015a	N/A	Duplicate of Fattore 2015
Giel 2017	N/A	Ineligible intervention/exposure: contract farming
Ismail 2020	Palm oil	Ineligible population: animal and cell studies
Jayawardena 2021	Coconut oil	Ineligible intervention/exposure: coconut oil
Jayawardena 2020	Coconut oil	Ineligible intervention/exposure: coconut oil
Koo 2006	Palm olein	Ineligible population: infants
Ma 2016	Coconut oil	Ineligible intervention/exposure: coconut oil
Mozaffarian 2009	Butter, lard and palm, cottonseed, high oleic sunflower, soybean and canola oil	Ineligible intervention/exposure: trans-fat replacement
Muhamad 2018	Palm oil	Ineligible outcomes: BMI, body weight, waist circumference
Myzabella 2019	Palm oil	Ineligible intervention/exposure: work in palm oil industry
Neelakantan 2020	Coconut oil	Ineligible intervention/exposure: coconut oil
Pupala 2019	Coconut oil	Ineligible intervention/exposure: coconut oil
Rabiee 2012	N/A	Ineligible population: animal study
Ryan 2020	N/A	Ineligible intervention/exposure: conservation marketing
Sun 2014	Palm oil	Duplicate of Sun 2015
Syarifah-Noratiqah 2020	Palm oil	Ineligible population: animal study
Teng 2020	Coconut oil	Ineligible intervention/exposure: coconut oil
Woolley 2020	N/A	Ineligible intervention/exposure: oil pulling
Yanza 2020	Medium chain fats	Ineligible population: animal study
Yu 2009	Palm oil	Ineligible population: infants
Ziegler 2012	N/A	Ineligible study design: agricultural/land-use study
Zulkiply 2019	Palm oil	Ineligible outcomes: glucose, HbA1c, homeostasis model assessment

BMI: body mass index; Hb: haemoglobin; N/A: not applicable.

References for Annex 1

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Annex 2

Details of included systematic reviews and methodological quality assessments

Table A2.1 Characteristics of included systematic reviews (*n*=5)

Study ID	Aim/objective	Eligible study designs	Eligible participants	Eligible comparison (treatment/ exposure and comparator)	Duration of intervention/ follow-up period of included studies	Prespecified outcomes	No. of studies and study designs included Countries	Funding declaration Conflict of interest statement
Hisham 2020	The study aimed to compare the effects of palm oil consumption with other edible oils rich in MUFAs and PUFAs on serum lipid profiles	Parallel group and cross-over RCTs	Not specified	Dietary intake of palm oil or palm olein compared to vegetable oils rich in MUFAs or PUFAs, or both	Interventions lasting more than 2 weeks were eligible for inclusion; durations ranged from 3 to 10 weeks	At least one of the following: TC, LDL-C, HDL-C, TG, Apo A1 and Apo B	A total of 23 trials were included: 3 parallel group and 20 cross-over RCTs; from 9 countries (Australia 3, China 2, Denmark 1, India 2, Malaysia 5, Netherlands 1, Sweden 1, Thailand 1, United States 7)	"The study was supported by the grant 'University Malaya Grand Challenge–SBS (Equitable Society)' (GC003E-17SBS)." "The authors declare that they have no conflicts of interest."

Ismail 2018	We aimed at systematically synthesizing the association of palm oil consumption with CVD risk and CVD-specific mortality, including specifically CHD and stroke	Parallel group, cluster and cross-over RCTs; quasi experiments; cohort, case–control and cross-sectional studies; case series	Adult participants (aged >18 years) with or without existing CVD	Comparison of palm oil intake with that of other types of vegetable oils, such as sunflower oil, canola oil, soybean oil or olive oil	Not specified in eligibility criteria; durations ranged from 4 to 18 years	Risk of CHD or stroke and cause-specific death rates (defined as the number of CHD or stroke deaths attributable to palm oil consumption)	A total of 4 studies were included: 3 case–control studies and 1 retrospective ecological study from 2 countries (Costa Rica 3, United States 1)	"The authors received no specific funding for this work." "The authors have declared that no competing interests exist."
Schwingshackl 2018	The aim of this NMA is to compare the effects of 13 different oils or solid fats on blood lipids	Parallel group and cross-over RCTs	Patients with a mean age ≥18 years	Diets varying in composition or isocaloric exchange of at least 2 of the following: safflower, sunflower, rapeseed, hempseed, flaxseed, corn, olive, soybean, palm and coconut oils; and lard, beef fat and butter	Eligible duration: at least 3 weeks; ranged from 3 to 27 weeks	LDL-C (primary NMA outcome) and TG (secondary NMA outcomes), TC, HDL-C	54 RCTs (16 parallel group, 36 cross-over and one unspecified RCTs) Australia 1, Brazil 2, Canada 1, China 1, Denmark 3, Finland 3, France 1, Germany 2, Greece 3, India 1, Iran 2, Italy 4, Japan 1, Malaysia 6, South Africa 1, Spain 4, Sweden 1, Switzerland 1, Thailand 1, United Kingdom 2, United States 13	"This work was supported by NutriAct–Competence Cluster Nutrition Research Berlin-Potsdam funded by the German Federal Ministry of Education and Research (FKZ: 01EA1408A-G). The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript." "The authors declare that they have no competing interests."

Study ID	Aim/objective	Eligible study designs	Eligible participants	Eligible comparison (treatment/ exposure and comparator)	Duration of intervention/ follow-up period of included studies	Prespecified outcomes	No. of studies and study designs included Countries	Funding declaration Conflict of interest statement
Sun 2015	We systematically reviewed the effect of palm oil consumption on blood lipids compared with other cooking oils using data from clinical trials	Clinical trials	Human subjects	Comparing the feeding of palm oil or palm olein, as cooking oil or cooked food, with feeding of any vegetable oil low in saturated fat, PHVO (trans-fat containing) or saturated animal fat	Feeding periods lasting for at least 2 weeks were eligible for inclusion; durations ranged from 2 to 16 weeks	Blood lipid concentrations such as TC, LDL-C (directly measured or calculated), HDL-C and TG	A total of 32 trials were included: 8 cross-over, 20 randomized cross-over, 1 parallel and 3 randomized parallel trials; from 13 countries (Australia 5, China 1, Denmark 1, India 1, Malaysia 8, Netherlands 1, Norway 1, South Africa 1, Spain 2, Thailand 1, United Kingdom 1, United States 8, Venezuela 1)	"Supported by the National Medical Research Council (NMRC 1270/2010), Singapore." "…no conflicts of interest."
Wang 2019	To assess the effects of palm oil consumption on plasma lipid profiles and CVD risk	RCTs	Subjects with normal serum cholesterol concentrations, without any metabolic syndrome or medical history of coronary artery disease and not taking any medications, such as oral contraceptives	Comparing diets rich in palm oil with diets rich in MUFA or PUFA	Intervention periods lasting for more than 2 weeks were eligible for inclusion; durations ranged from 4 to 8 weeks	TC, LDL-C (primary review outcomes), HDL-C and TG (secondary review outcomes)	A total of 11 trials were included: 2 cross-over, 5 randomized cross-over, 3 randomized parallel trials and 1 before-after study; from 4 countries (China 3, India 1, Malaysia 6, United States 1)	No funding declaration. "None of the authors declared a conflict of interest."

Apo: apolipoprotein; CHD: coronary heart disease; CVD: cardiovascular disease; HDL-C: high-density lipoprotein cholesterol; ID: identification; LDL-C: low-density lipoprotein cholesterol; MUFA: monounsaturated fatty acid; NMA: network meta-analysis; PHVO: partially hydrogenated vegetable oils; PUFA: polyunsaturated fatty acid; RCT: randomized controlled trial; TC: total cholesterol; TG: triglyceride; United Kingdom: United Kingdom of Great Britain and Northern Ireland.

Note: This table summarizes the characteristics for the full systematic review, not only the characteristics relevant to the data extracted for this scoping review.

Table A2.2 Interpreting the AMSTAR 2 ratings of overall confidence in the results of a systematic review

High	No or one non-critical weakness: the systematic review provides an accurate and comprehensive summary of the results of the available studies that address the question of interest
Moderate	More than one non-critical weakness:[a] the systematic review has more than one weakness but no critical flaws. It may provide an accurate summary of the results of the available studies that were included in the review
Low	One critical flaw with or without non-critical weaknesses: the review has a critical flaw and may not provide an accurate and comprehensive summary of the available studies that address the question of interest
Critically low	More than one critical flaw with or without non-critical weaknesses: the review has more than one critical flaw and should not be relied on to provide an accurate and comprehensive summary of the available studies

[a] Multiple non-critical weaknesses may diminish confidence in the review and it may be appropriate to move the overall appraisal down from moderate to low confidence.

Source: Shea 2020.

Table A2.3 AMSTAR 2 critical domains, regarded as those that critically affect the validity of a review and its conclusions

Domain	Description
2	Protocol registered before commencement of the review
4	Adequacy of the literature search
7	Justification for excluding individual studies
9	Risk of bias from individual studies being included in the review
11	Appropriateness of meta-analytical methods
13	Consideration of risk of bias when interpreting the results of the review
15	Assessment of presence and likely impact of publication bias

Table A2.4 AMSTAR 2 judgements per included systematic review

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
1. Did the research questions and inclusion criteria for the review include the components of PICO?							
No	• Intervention • Comparator group • Outcome • Timeframe for follow-up	Yes	• Population • Intervention • Comparator group • Outcome	Yes	• Population • Intervention • Comparator group • Outcome • Timeframe for follow-up	Yes	• Population • Intervention • Comparator group • Outcome • Timeframe for follow-up
2. Did the report of the review contain an explicit statement that the review methods were established before the conduct of the review and did the report justify any significant deviations from the protocol?							
No	N/A	No	• Review question(s) • A search strategy • A RoB assessment • A meta-analysis or synthesis plan	No	• Review question(s) • A search strategy • Inclusion or exclusion criteria • A meta-analysis or synthesis plan • A plan for investigating causes of heterogeneity	No	N/A
3. Did the review authors explain their selection of the study designs for inclusion in the review?							
No	N/A	No	N/A	No	N/A	No	N/A

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
4. Did the review authors use a comprehensive literature search strategy?							
Partial yes	• Searched at least 2 databases • Provided keyword and/or search strategy • Justified publication restrictions • Searched the reference lists/ bibliographies of included studies • Searched trial/study registries • Conducted the search within 24 months of completion of the review	Partial yes	• Searched at least 2 databases • Provided keyword and/or search strategy • Justified publication restrictions • Searched the reference lists or bibliographies of included studies • Searched trial/study registries • Conducted the search within 24 months of completion of the review	Partial yes	• Searched at least 2 databases • Provided keyword and/or search strategy • Justified publication restrictions • Searched the reference lists/ bibliographies of included studies • Conducted the search within 24 months of completion of the review	No	• Searched at least 2 databases • Provided keyword and/or search strategy • Searched the reference lists/ bibliographies of included studies • Conducted the search within 24 months of completion of the review
5. Did the review authors perform study selection in duplicate?							
Yes	• At least 2 reviewers independently agreed on selection of eligible studies and achieved consensus on which studies to include	Yes	• At least 2 reviewers independently agreed on selection of eligible studies and achieved consensus on which studies to include	Yes	• At least 2 reviewers independently agreed on selection of eligible studies and achieved consensus on which studies to include	Yes	• At least 2 reviewers independently agreed on selection of eligible studies and achieved consensus on which studies to include
6. Did the review authors perform data extraction in duplicate?							
Yes	• At least 2 reviewers achieved consensus on which data to extract from included studies	Yes	• At least 2 reviewers achieved consensus on which data to extract from included studies	Yes	• At least 2 reviewers achieved consensus on which data to extract from included studies	Yes	• At least 2 reviewers achieved consensus on which data to extract from included studies

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
7. Did the review authors provide a list of excluded studies and justify the exclusions?							
Yes	• Provided a list of potentially relevant studies that were read in full-text form but excluded from the review • Justified the exclusion from the review of each potentially relevant study	No	N/A	No	N/A	Yes	• Provided a list of potentially relevant studies that were read in full-text form but excluded from the review • Justified the exclusion from the review of each potentially relevant study
8. Did the review authors describe the included studies in adequate detail?							
Partial yes	• Populations • Interventions • Comparators • Outcomes • Research designs • Population in detail • Intervention in detail • Comparator in detail • Timeframe for follow-up	Partial yes	• Populations • Interventions • Comparators • Outcomes • Research designs • Intervention in detail • Comparator in detail • Timeframe for follow-up	No	• Populations • Interventions • Comparators • Research designs • Population in detail • Intervention in detail • Comparator in detail • Timeframe for follow-up	No	• Populations • Interventions • Comparators • Research designs • Intervention in detail • Comparator in detail • Timeframe for follow-up
9. Did the review authors use a satisfactory technique for assessing the RoB in individual studies that were included in the review?							
RCTs	RCTs	RCTs	RCTs				
Yes	• Unconcealed allocation • Lack of blinding of patients and assessors when assessing outcomes • Allocation sequence that was not truly random • Selection of the reported result from among multiple measurements or analyses of a specified outcome	Includes only NRSIs	N/A	No	• Lack of blinding of patients and assessors when assessing outcomes • Allocation sequence that was not truly random	Yes	• Unconcealed allocation • Lack of blinding of patients and assessors when assessing outcomes • Allocation sequence that was not truly random • Selection of the reported result from among multiple measurements or analyses of a specified outcome

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
NRSIs	NRSIs	NRSIs	NRSIs				
Includes only RCTs	N/A	Partial yes	• From confounding • From selection bias • Methods used to ascertain exposures and outcomes	No	• From selection bias	No	• From selection bias • Selection of the reported result from among multiple measurements or analyses of a specified outcome
10. Did the review authors report on the sources of funding for the studies included in the review?							
Yes	• Must have reported the sources of funding for individual studies included in the review	No	N/A	Yes	• Must have reported the sources of funding for individual studies included in the review	Yes	• Must have reported the sources of funding for individual studies included in the review
11. If meta-analysis was performed did the review authors use appropriate methods for statistical combination of results?							
RCTs	RCTs	RCTs	RCTs				
Yes	• The authors justified combining the data in meta-analysis • The authors used an appropriate weighted technique to combine study results and adjusted for heterogeneity if present • The authors investigated the causes of any heterogeneity	No meta-analysis conducted	N/A	Yes	• The authors justified combining the data in meta-analysis • The authors used an appropriate weighted technique to combine study results and adjusted for heterogeneity if present • The authors investigated the causes of any heterogeneity	Yes	• The authors justified combining the data in meta-analysis • The authors used an appropriate weighted technique to combine study results and adjusted for heterogeneity if present • The authors investigated the causes of any heterogeneity
NRSIs	NRSIs	NRSIs	NRSIs				
No meta-analysis conducted	N/A	No meta-analysis conducted	N/A	No	N/A	No	N/A

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
12. If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?							
Yes	• The pooled estimate was based on RCTs at variable RoB, authors performed analyses to investigate possible impact of RoB on summary estimates of effect	No meta-analysis conducted	N/A	Yes	• The pooled estimate was based on RCTs and NRSIs at variable RoB, authors performed analyses to investigate possible impact of RoB on summary estimates of effect	No	N/A
13. Did the review authors account for RoB in individual studies when interpreting or discussing the results of the review?							
No	N/A	Yes	• NRSIs with moderate or high RoB were included, the review provided a discussion of the likely impact of RoB on the results	No	N/A	No	N/A
14. Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review?							
No	N/A	Yes	• There was no significant heterogeneity in the results	Yes	• Heterogeneity was present, authors performed an investigation of sources of any heterogeneity in the results and discussed the impact of this on the results of the review	Yes	• Heterogeneity was present, authors investigated sources of any heterogeneity in the results and discussed the impact of this on the results of the review
15. If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review?							
Yes	• Performed graphical or statistical tests for publication bias and discussed the likelihood and magnitude of impact of publication bias	No meta-analysis conducted	N/A	Yes	• Performed graphical or statistical tests for publication bias and discussed the likelihood and magnitude of impact of publication bias	No	N/A

Hisham 2020		Ismail 2018		Sun 2015		Wang 2019	
Decision	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**	**Decision**	**Elements included**
16. Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review?							
Yes	• The authors reported no competing interests	Yes	• The authors reported no competing interests	Yes	• The authors reported no competing interests	Yes	• The authors reported no competing interests
OVERALL JUDGEMENT							
Hisham 2020 is a low-quality review		**Ismail 2018 is a critically low-quality review**		**Sun 2015 is a critically low-quality review**		**Wang 2019 is a critically low-quality review**	

N/A: not applicable; NRSI: non-randomized studies of interventions; PICO: population, intervention, comparator and outcome; RCT: randomized controlled trial; RoB: risk of bias; systematic review.

Table A2.5 ISPOR-AMCP-NPC Good Practice Task Force checklist[a] judgements for included systematic review with NMA

	Schwingshackl 2018	
Evidence base used for the indirect comparison or NMA	1.	Did the researchers attempt to identify and include all relevant RCTs?
		Strength
	2.	Do the trials for the interventions of interest form one connected network of RCTs?
		Strength
	3.	Is it apparent that poor quality studies were included thereby leading to bias?
		Can't answer (included RCTs were mostly at low or unclear risk of bias, with a minority of studies having high risk of bias relating to performance and attrition bias)
	4.	Is it likely that bias was induced by selective reporting of outcomes in the studies?
		Can't answer (included RCTs were mostly at low risk of reporting bias, with a minority of studies having unclear risk of reporting bias)
	5.	Are there systematic differences in treatment effect modifiers (i.e. baseline patient or study characteristics that impact the treatment effects) across the different treatment comparisons in the network?
		Weakness
	6.	If yes (i.e. there are such systematic differences in treatment effect modifiers), were these imbalances in effect modifiers across the different treatment comparisons identified before comparing individual study results?
		Strength
	OVERALL DOMAIN JUDGEMENT	
	Neutral	
Analysis	7.	Were statistical methods used that preserve within-study randomization? (No naive comparisons)
		Strength
	8.	If both direct and indirect comparisons are available for pairwise contrasts (i.e. closed loops), was agreement in treatment effects (i.e. consistency) evaluated or discussed?
		Strength
	9.	In the presence of consistency between direct and indirect comparisons, were both direct and indirect evidence included in the NMA?
		Strength
	10.	With inconsistency or an imbalance in the distribution of treatment effect modifiers across the different types of comparisons in the network of trials, did the researchers attempt to minimize this bias with the analysis?
		Weakness
	11.	Was a valid rationale provided for the use of random effects or fixed effect models?
		Can't answer (not reported; the ISPOR checklist advocates for random effects models for clinically or methodologically diverse studies)
	12.	If a random effects model was used, were assumptions about heterogeneity explored or discussed?
		Weakness
	13.	If there were indications of heterogeneity, were subgroup analyses or meta-regression analysis with prespecified covariates performed?
		Weakness
	OVERALL DOMAIN JUDGEMENT	
	Weakness	

	Schwingshackl 2018
Reporting quality and transparency	14. Is a graphical or tabular representation of the evidence network provided with information on the number of RCTs per direct comparison?
	Strength
	15. Are the individual study results reported?
	Strength
	16. Are results of direct comparisons reported separately from results of the indirect comparisons or NMA?
	Weakness
	17. Are all pairwise contrasts between interventions as obtained with the NMA reported along with measures of uncertainty?
	Strength
	18. Is a ranking of interventions provided given the reported treatment effects and its uncertainty by outcome?
	Strength
	19. Is the impact of important patient characteristics on treatment effects reported?
	Weakness
	OVERALL DOMAIN JUDGEMENT
	Neutral
Interpretation	20. Are the conclusions fair and balanced?
	Strength
	OVERALL DOMAIN JUDGEMENT
	Strength
Conflict of interest	21. Were there any potential conflicts of interest?
	Strength
	22. If yes, were steps taken to address these?
	N/A
	OVERALL DOMAIN JUDGEMENT
	Strength
	IS THE CREDIBILITY OF THE NMA SUFFICIENT OR INSUFFICIENT?
	Sufficient

N/A: not applicable; NMA: network meta-analysis; RCT: randomized controlled trial.

[a] Jansen 2014.

References for Annex 2

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